



System-of-systems safety for low-altitude aviation transportation

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ABSTRACT

Low-altitude (LA) transportation is rapidly evolving as a new sector of traditional aviation industry. Specifically, existing aviation safety studies were largely developed under traditional aviation System-of-systems (SoS) architectures, and primarily address system-level risks (human, vehicle, communication-navigation-surveillance, or operational management) with clear system boundaries. As a new SoS, LA SoS may have new emergent risk types and cross-domain cascading risk propagation via new architecture. This paper considers the LA transportation SoS from architecture viewpoint, and analyzes its new characteristics and risk patterns. Therefore, we propose LA SoS safety engineering framework to handle these possible new challenges through architecture design, testing & evaluation, and safety management. Accordingly, our SoS safety engineering framework mainly includes: architecture design (human-automation safety control and digital flight rules), testing & evaluation (traceable design-to-verification and equipment safety requirement), and safety management (monitor-assess-mitigate for in-time risk prediction and mitigation). Our safety engineering framework aims to provide possible solution for this new complex system of LA aviation transportation.

1. Introduction

The low-altitude (LA) economy represents a holistic economic paradigm propelled by various LA flight activities, encompassing both manned and unmanned operations, which in turn fosters integrated growth across related sectors. Over the past decade, different countries have introduced innovative concepts and development strategies aimed at advancing the LA economy [1–3]. LA transportation, as the “third dimension” of urban mobility, is progressively moving toward a practical reality. Future LA transportation scenarios will need to accommodate a vast number of flight requests. One study analyzing population and transportation data from 1,200 cities worldwide estimates that by 2050, 160,000 passenger-class autonomous aircraft will be in use worldwide [4]. Other research suggests that, in an optimal scenario, demand for air taxis and airport shuttle services could reach 11 million daily trips, accounting for 20% of all daily commutes in the U.S. [5]. Some forecasts predict up to 87,000 drones per hour could be flying within the urban airspace of Paris by 2035 [6].

However, large-scale Urban Air Mobility (UAM) deployment confronts multifaceted challenges in safety, noise, energy efficiency, and

infrastructure, as identified by EASA [7], NASA [8], and the FAA [9]. Public acceptance is hindered by environmental, privacy, and social equity concerns, with noise pollution posing risks to health and quality of life [10]. Moreover, eVTOLs remain energy-intensive and cost-prohibitive for applications, undermining their economic viability compared to ground transport [11–13]. Among these issues, safety is a fundamental requirement for LA operations. Current research on LA safety primarily focuses on mitigating risks from isolated components, such as aircraft design, communication-navigation-surveillance (CNS), and conflict management [14,15]. These efforts aim to address known hazards in specific subsystems or operational phases. Nevertheless, the LA transportation is inherently evolving into a new type of complex system-of-systems (SoS), distinct from conventional transportation frameworks [16]. In such an environment, even if individual components or single-aspect safety measures are adequately assured, real-world operations may introduce unforeseen risks arising from complex interactions among different stakeholders. This entails the necessity to consider the LA transportation as a new SoS, analyze new characteristics and risk patterns from the SoS perspective and propose new safety engineering framework for LA SoS.

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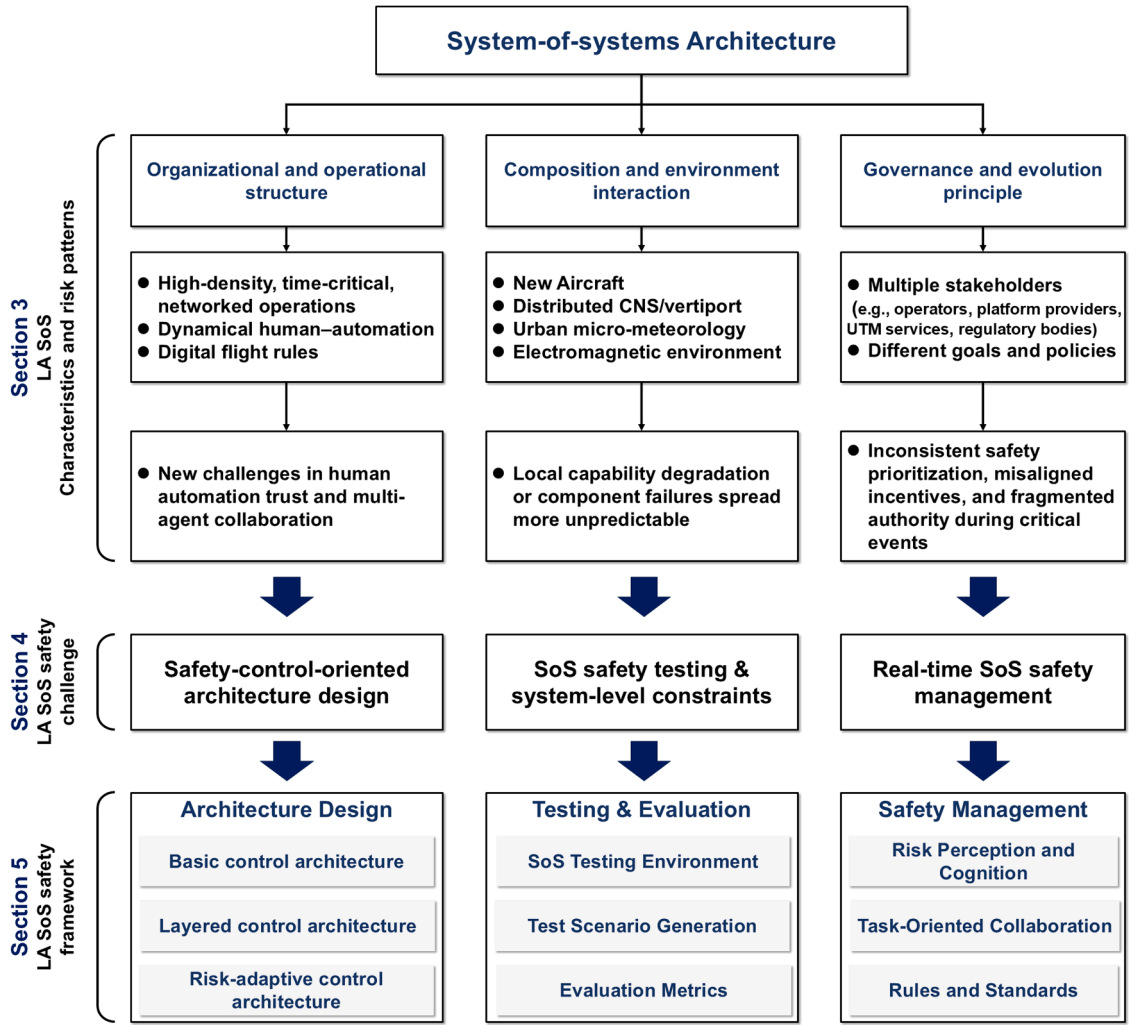


Fig. 1. Logical thread of this review.

As illustrated in Fig. 1, this paper considers the LA transportation as a new SoS, using the ISO/IEC/IEEE 42010 architecture lens: (i) composition and environment interaction, (ii) organizational and operational structure, and (iii) governance and evolution principle. Following logic of “new risk patterns→safety challenge→possible SoS safety engineering framework”, we propose LA SoS safety engineering framework to handle possible new challenges through architecture design, testing & evaluation, and safety management.

- (1) For organizational and operational structure risks, as LA transportation is shifting beyond the conventional aviation SoS

architecture, it needs to design a new LA SoS architecture with reliable human-automation safety control and digital flight rule. Human-automation safety control loop is the basic unit of LA SoS safety, which decides how fast and reliable an operational risk can be controlled. Trustable artificial intelligence (AI) will be deeply embedded to assure its reliability. Digital flight rules are a set of automated regulations which will coordinate LA traffic to maintain high safety and efficiency.

- (2) For composition and environment interaction risks, we handle composition safety through SoS safety testing and system safety requirement identification. We derive SoS-level safety

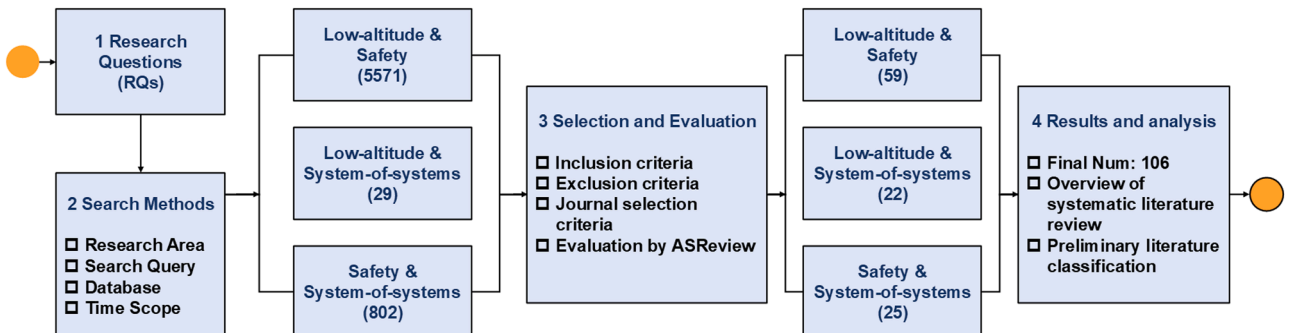


Fig. 2. Research methodology of the systematic literature review.

Table 1

Research questions.

No.	Research Questions
RQ1	What new SoS characteristics and systemic risk patterns arise in LA scenarios compared with conventional civil aviation?
RQ2	To what extent are existing studies adequate for addressing the safety challenges of LA SoS?
RQ3	What SoS architecture and management are needed for LA SoS safety?

Table 2

Search query and scope for cross-domains.

No.	Research Area	Search Query	Database	Time Scope
CD1	LA & Safety	TS = (urban air mobility OR low altitude OR UAM) AND TS = (safety OR risk)	WoS Core Collection	2021-present
CD2	LA & SoS	TS = ("system of systems" OR "systems of systems") AND TS = (urban air mobility OR low altitude OR UAM)	WoS Core Collection	1985-present
CD3	SoS & Safety	TS = ("system of systems" OR "systems of systems") AND TS = (safety OR risk)	WoS Core Collection	1985-present

requirements and decompose them into system-level constraints (including airworthiness requirements) to ensure composition safety. In testing & evaluation, we validate these requirements and safety interactions with environment in digital-physical combined test platform via scenario-based evaluation.

- (3) For governance and evolution principle risks, the framework extends traditional organization-centric Safety Management System (SMS) towards In-Time Aviation SMS (e.g. IASMS) [135]. Traditional SMS is largely post-event and labor-intensive, which makes it difficult to detect and mitigate evolving risks. In contrast, IASMS extends SMS by integrating in-time safety assurance with traditional risk management experience and provides a distributed, federated “monitor-assess-mitigate” decision loop that fuses cross-domain data to predict risk and trigger mitigation in time.

The structure of the paper is outlined as follows. Section 2 introduces the research methodology and results for the literature review. Section 3 summarizes new characteristics and associated risks from the SoS perspective. Section 4 analyzes new challenges by comparing traditional aviation practices and existing SoS safety research. Section 5 presents a possible SoS safety engineering framework as the corresponding solution. Section 6 concludes the paper.

2. Research methodology and results

In our review, we provide a literature review on the intersections of LA, SoS, and safety. It followed the semi-automatic systematic literature review guidelines outlined by van Dinter et al. [17], ensuring reproducibility and minimizing bias [18]. The methodology for conducting the systematic literature review is illustrated in Fig. 2 and includes the following stages: research questions (RQs), search methods, selection & evaluation, and results & analysis.

Table 3

Study inclusion and exclusion criteria.

No.	Criteria	CD1	CD2	CD3
IC1	The study must have full text (e.g., abstract-only papers are not considered)			
IC2	The study must be a primary study			
IC3	/		The studies must be published in journals classified under the Journal selection criteria 2 and Journal selection criteria 3 category in Table 4	The studies must be published in journals classified under the Journal selection criteria 1 and Journal selection criteria 2 category in Table 4
IC4	The study must be directly relevant to safety aspects within the LA domain		The study must adopt an SoS perspective to investigate LA operations or scenarios	The study must be safety-related and explicitly focus on an SoS as its primary subject
EC1	The study is duplicated			
EC2	The study belongs to non-English language publications			

2.1. Research questions

Corresponding to the contributions outlined in Section 1, the core RQs addressed in this paper are listed in Table 1.

2.2. Search methods

The literature selection process aimed to capture research from three cross-domain (CD) perspectives, focusing on the intersection of LA, SoS and safety. As shown in Table 2, we report the exact search strings/keywords, the databases and sources searched, including Web of Science (WoS) core collection, and the corresponding time windows.

2.3. Selection and evaluation

For selection criteria, as outlined in Table 3, we provide the inclusion criteria (IC), exclusion criteria (EC) and journal selection criteria (JSC) for all three CDs. The evaluation of the selected studies, or the second round of screening, was performed using a data-driven systematic literature review automation tool (ASReview [17]). In this phase, not only studies analyzing LA risks but also those proposing solutions to the challenges faced by LA SoS were included to ensure a comprehensive review of the field.

2.4. Results and analysis

We present the number of records at each stage (identified, after applying selection criteria, after evaluation, after deduplication, and final included set) and describe how the final set of studies was reviewed and synthesized, including the categorization scheme used to organize the evidence. As shown in Table 5, the initial search across the three intersecting research directions yielded a literature volume of 6,402 publications, which was refined to a final set of 106 after applying the screening and evaluation criteria. To focus on safety research, our

Table 4
Journal selection criteria for different area.

No.	Research Area	Journal selection
JSC1	Safety & Risk	Reliability Engineering & System Safety IEEE Transactions on Reliability Safety Science Risk Analysis Engineering
JSC2	System/SoS Engineering	IEEE Transactions on Systems, Man, and Cybernetics: Systems IEEE Systems Journal IEEE Transactions on Engineering Management Systems Engineering Progress in Aerospace Sciences Sustainable Cities and Society Aerospace Science and Technology
JSC3	Aerospace	IEEE Transactions on Aerospace and Electronic Systems Drones Aerospace

Table 5
Overview of systematic literature review.

No.	Automated search	After selection	After evaluation	Final selected studies
CD1	5571	168	59	[19–77]
CD2	29	29	22	[78–99]
CD3	802	79	25	[100–124]

selection criteria included a predefined list of target journals (see Table 4) to ensure the quality and relevance of the included studies for safety analysis. Therefore, a large number of records were excluded during the selection stage, especially those in CD1 and CD3.

To systematically organize the knowledge domain of LA SoS safety, the ultimately selected publications were classified, with the preliminary categorization results presented in Table 6. The classification is primarily based on three intersecting research directions: (i) LA & Safety: This direction examines safety specifically within the LA domain. (ii) SoS & Safety: This direction addresses theories, methods, and challenges related to the safety of Systems-of-Systems (SoS), not specific to the LA domain. (iii) LA & SoS: This direction focuses on the architectural design and operational management of SoS within the LA domain.

3. LA as a new SoS

ISO 21839:2019 describes an SoS as a set of interacting systems that provide unique capabilities unattainable by any single constituent system [125]. There are various definitions and characteristics in the research of SoS. For example, Maier summarizes SoS characteristics into five properties (“Maier’s criteria”) [126]: (i) operational independence, (ii) managerial independence, (iii) geographic distribution, (iv) emergent behavior, and (v) evolutionary development. Moreover, Boardman and Sauser distinguish SoS from conventional systems using five identifying characteristics—Autonomy, Belonging, Connectivity, Diversity, and Emergence [127], while DeLaurentis propose a taxonomy along three orthogonal dimensions, namely Connectivity, Control/Autonomy, and System Type, to characterize and compare different SoS forms [128]. In this review, we describe LA SoS according to standards of ISO/IEC/IEEE 42010:2020 [129]. ISO/IEC/IEEE 42010:2020 states that an entity’s architecture includes “its constituent elements, interactions among elements, interactions with the environment, as well as its behavior/structure and the principles governing its design, use, operation and evolution.” Accordingly, we group the above as three architecture aspects:

Table 6
Preliminary literature classification.

Cross-domain	Categories	Main focuses
LA & Safety	Platform/System	The safety of individual systems or platforms within the LA architecture (e.g., Unmanned Aircraft Vehicle (UAV) platforms, flight control systems, and Sense and Avoid systems).
	Interaction/Network	Ensuring safe interactions among multiple systems connected by communication and cooperation networks (e.g., UAV-to-UAV, UAV-to-Infrastructure, and underlying communication & connectivity networks).
	Operation/Management	Ensuring the overall safety of large-scale operations through airspace design, traffic management, and risk assessment (e.g., airspace and route network design, risk assessment and management, traffic flow and conflict management).
LA & SoS	Top-Level Design & Framework	The overall architecture, conceptual design, SoS integration, standards, and the overall resilience of the LA system to disruptions (e.g., off-nominal events).
	Operations & Management	The specific operational aspects of the LA system, including airspace and route network design, traffic management, risk assessment, fleet management, and analysis of operational limits.
	Modeling, Simulation, and Analytical Tools & Methods	The underlying technical tools, simulation platforms, data analysis methods, and frameworks that support the design, analysis, and evaluation of the LA system.
SoS & Safety	Theoretical Frameworks	The overall safety philosophy, fundamental theoretical frameworks, general guiding principles, and macro-level challenges of SoS safety, providing a methodological foundation.
	Risk Assessment & Management Methods	Specific risk assessment, modeling, management tools, and techniques for System of Systems, including data-driven analysis, simulation, and optimization frameworks, for quantifying and managing systemic risks.
	Application Cases & Empirical Studies	Applying SoS safety theories and methods to specific domains (e.g., aviation, nuclear power, critical infrastructure), validating their effectiveness through case studies.

- Composition and environment interaction (elements, relationships, and their coupling with the environment)
- Organizational and operational structure (structure and behavior)
- Governance and evolution principle (design, operation, and management)

Specifically, by considering the LA transportation system as a new SoS, this section analyzes new characteristics and risk patterns compared with conventional civil aviation: (i) Composition and environment interaction, where in LA SoS new aircraft and distributed CNS/vertiport infrastructure operate under unique urban micro-meteorology and electromagnetic environment, making possible local capability degradation or component failures spread more unpredictable; (ii) Organizational and operational structure, where high-density, time-critical, networked operations require dynamical human-automation function allocation and digital flight rules, leading to new challenges

in human-automation trust and multi-agent collaboration; and (iii) Governance and evolution principle, where multiple stakeholders (e.g., operators, platform providers, Unmanned Aircraft System Traffic Management (UTM) services, regulatory bodies) may pursue different goals under different policies, possibly leading to inconsistent safety prioritization, misaligned incentives, and fragmented authority during critical events.

3.1. Composition and environment interaction

3.1.1. New characteristics

From the SoS architecture aspect for composition and environment interaction, new aircraft and distributed CNS/vertiport infrastructure in LA SoS operate under unique urban micro-meteorology and electromagnetic environment. Supporting literature indicates that this “composition-environment coupling” in LA SoS is qualitatively different from traditional civil aviation in three aligned ways.

- (1) New LA aircraft and distributed CNS/vertiport infrastructure are becoming core building blocks of the LA SoS. Following NASA's Revolutionary Vertical Lift Technology typology [130], LA aircraft can be described by three attributes: Vertical Take-Off and Landing configuration and rotor count, propulsion and energy architecture, and primary control scheme. These new aircraft are inherently environment-coupled in the built environment, and recent aircraft-design reviews and SoS-based aircraft concept assessments reflect this coupling when evaluating aircraft concepts under ecosystem-level constraints [65,87]. Urban CNS is also environment-coupled. Global-navigation-satellite-system integrity monitoring and redundant inertial sensing mitigate multipath and occlusion effects and improve robustness to low-cost sensor faults [53,68]. Communication reliability, including latency, coverage gaps, and handovers, also affects coordination [56,64]. Vertiports are safety-critical ground infrastructure, and their design, capacity, operations, and regulation can constrain system performance and risk [73].
- (2) Urban micro-meteorology (e.g. near-building turbulence, wakes, and wind shear) in the built environment drives aircraft-environment interaction uncertainty in LA SoS. A recent review synthesizes how building-induced flow complexity and urban thermal effects can directly threaten UAM takeoff/landing/hover stability and even interact with rotor downwash in ways that remain insufficiently quantified [39]. Some dedicated studies further show that urban wind fields measurably alter LA aircraft performance and control burden [23]. This makes “urban micro-meteorology-aware” important interaction concern, motivating approaches such as UAM-oriented wind-field sensing using airborne Doppler light detection and ranging [20] and refined urban geometry representations that preserve local flow-relevant details [21].
- (3) Urban electromagnetic complexity strongly shapes CNS performance in LA SoS. Global-navigation-satellite-system multipath and Non-Line-of-Sight receptions drive integrity challenges and motivate environment-aware protection-level estimation [53], while redundant inertial sensing architectures are proposed to mitigate low-cost sensor vulnerability and integrity degradation [68]. In parallel, UAM communications and networking surveys argue that coverage/latency/reliability requirements become binding constraints in dense urban deployment, reinforcing the need to treat communication connectivity as part of the SoS architecture [56]. Finally, because these elements are coupled, some works explicitly study aircraft/SoS co-design and show that vehicle concepts and fleet performance are sensitive to operational context and infrastructure assumptions in a way that

cannot be extrapolated from traditional civil aviation baselines [81,91,92].

3.1.2. Risk patterns

We place system-level risk sources (e.g., new aircraft or distributed CNS/vertiport degradation/failures) in this category, and link them with environment interaction of unique urban micro-meteorology and electromagnetic environment, possibly resulting in SoS risk chains such as “system degradation under unprecedented environmental disturbance→unidentified risk signals→biased situation awareness/decisions→reduced safety margins.” The reviewed literature supports this risk-chain logic by showing how small disturbances become systemically consequential in LA, city-embedded operations.

- (1) Urban micro-meteorology can be a primary driver of performance degradation, where building-induced turbulence, thermal buoyancy, and wind shear elevate track deviation and control workload. This is repeatedly identified as a core safety threat for LA phases in urban settings [39], which is consistent with empirical/analytical findings that urban wind fields can significantly perturb UAV flight dynamics [23]. Such disturbances directly compress separation margins in constrained geometries. This motivates designs that explicitly account for wind- and Global-Positioning-System-driven deviations when defining airspace blocks or separation assumptions [74]. It also motivates safety-assured guidance formulations that bound collision probability under uncertainty, rather than assuming deterministic separation [88].
- (2) Urban electromagnetic environment and built-environment occlusions can erode CNS integrity and create “unidentified risk signals”. Environment-aware global navigation satellite system integrity monitoring shows that protection levels and availability can change sharply across suburban, urban, and urban-canyon contexts [53]. To mitigate such integrity degradation, redundancy-based inertial integration is proposed to prevent low-cost sensor faults from silently degrading navigation integrity [68]. At the tactical layer, non-cooperative detect-and-avoid depends on multi-sensor perception whose failure modes are environment-coupled. Radar and visual fusion demonstrates how clutter and detection uncertainty at LA can be reduced at the fusion stage [30]. Lidar-based corridor detect-and-avoid further emphasizes that intruder sensing and avoidance often needs to run in real time under corridor constraints [71].
- (3) Constrained urban airspace plus ground exposure turns deviations into cascading conflict and harm potential. Collision-risk-based separation demarcation for small UAV encounters shows sensitivity to encounter geometry and error bounds [25]. Structured airspace concepts (lanes, layers, and corridors) are introduced partly to prevent conflict amplification and reduce scheduling/tactical complexity under contingencies [69,74]. Meanwhile, urban ground-risk mapping and distribution estimation frameworks demonstrate that off-nominal events in LA operations have direct third-party risk implications (people/-vehicles/property), and that risk is spatially heterogeneous and strongly shaped by urban features [41,42].

3.2. Organizational and operational structure

3.2.1. New characteristics

From the SoS architecture aspect of organizational and operational structure, high-density, time-critical operations require dynamical human-automation function allocation and digital flight rules. Supporting literature indicates that the LA SoS operational-organizational structure departs from traditional civil aviation in three aligned ways.

- (1) LA operations are shifting from controller-centric control to scalable, automation-heavy traffic services. To scale conflict management in such services, it is often framed as a strategic-tactical layered process, consistent with ICAO Doc 9854 [131], which distinguishes strategic conflict management from tactical separation provision. Accordingly, lane- and graph-based large-scale UTM concepts combine strategic demand-capacity balancing with learning-based tactical deconfliction [55], and hybrid AI-enabled four-dimensional trajectory management performs real-time demand-capacity balancing and dynamic replanning under dense LA loads [57]. At the tactical layer, a wide set of multi-agent conflict detection/resolution and collision-avoidance mechanisms are explicitly designed [28,34,58,72,76]. Unlike conventional enroute operations, centralized procedures usually preserve time margins and separation buffers. In LA operations, conflicts often arise with tight reaction windows and constrained geometry, which motivates on-the-fly maneuver coordination and real-time separation assurance [51].
- (2) High-density LA operations force dynamic human-automation function allocation for scalable oversight. Scalable LA operations require simultaneously (i) high degrees of onboard autonomy, (ii) ground-based service automation (UTM/U-space-like), and (iii) human oversight under workload and accountability constraints. Reviews and frameworks emphasize that as automation increases, humans shift from direct separation provision toward supervision and exception handling. The transition itself is commonly treated as a safety-critical design and certification concern [38,90]. This is complemented by human-machine interface studies for monitoring multiple small unmanned aircraft system in urban operations, which highlight that sustained dense operations create information-overload risks and therefore demand interface designs that support attention guidance and anomaly salience [75].
- (3) Digital flight rules become a core operational primitive for scalable LA separation and access control. SoS need to encode separation, priority, access, and airspace-structure constraints into machine-executable rules to enable consistent, scalable coordination. This is reflected in risk-threshold separation demarcation that aligns rule parameters with target/equivalent level of safety principles [49]. It also appears in “digital traffic light” schemes that formalize controlled conflict regions and broadcast unified avoidance instructions [46]. Layered and structured airspace concepts use rule-encoded geometry as a mechanism to jointly support safety and efficiency at scale [62,77]. Moreover, multiple works show that the choice of what information is shared and permitted (e.g., intent, privacy-limited data, remote identification ecosystems, or permit assignment) becomes part of the rule system that shapes achievable trade-offs between safety and efficiency [31,36,67].

3.2.2. Risk patterns

We emphasize cross-system cascading risks induced by high-density, time-critical operations, which require dynamical human-automation function allocation and digital flight rules. When new conflict situation emerges that has never been trained in artificial intelligence model, decision uncertainty may increase in automated pilot and be amplified by cascading disruptions among multi-agents of different aircraft—thereby raising new challenges in human-automation trust and multi-agent collaboration under safety-critical decision-making. The reviewed literature supports this risk-chain logic by showing how tight coupling across traffic services, automation, and stakeholders can transform “local off-nominal events” into SoS-level disruptions.

- (1) High-density time-critical operations create conflict-amplification and domino risks. Under this operational scenario, a small deviation or local loss of separation may trigger

sequential maneuver interactions and secondary conflicts. Online coordination studies explicitly target “domino” separation-loss phenomena in congested airspace and show why coordination policies (often rule-constrained) are required to prevent conflict propagation [51]. Similarly, integrated strategic-tactical conflict management highlights that if density is not pre-conditioned (e.g., via demand-capacity balancing), tactical learning-based resolver may fail to maintain target safety levels under overload [55,57]. Multi-agent collision avoidance and tactical deconfliction works further imply that the operational hazard is not only pairwise collision risk but multi-agent coupling—i.e., resolution actions interact, producing emergent conflict chains when the encounter set grows or becomes heterogeneous [28,34,58,72,76].

- (2) Automation uncertainty and dynamical function allocation introduces “hidden brittleness → trust/coordination failure” pathways. Reviews of advanced air mobility stress that inserting AI into safety-critical separation/traffic functions demands rigorous assurance and certification. Otherwise, generalization gaps and limited explainability can yield brittle decision-making exactly in rare/untrained conflicts [38]. This concern is operationalized by stress-testing frameworks that automatically discover boundary failure modes of UTM decision systems in simulation [95]. Beyond algorithmic brittleness, these pathways also depend on whether human supervisors can perceive and act on emerging anomalies in time. Accordingly, human-machine interface guidance for monitoring multiple vehicles addresses the risk that automation and dense displays overwhelm supervisors, leading to missed anomalies or delayed interventions [75].
- (3) Networked dependencies across services and infrastructure can turn local performance degradations into systemic safety-margin loss. LA operations rely on interdependent service chains, so a degradation in one element can propagate across layers and erode safety margins. Surveys of UAM communications and networking further note that insufficient connectivity can reduce shared situational awareness and coordination, increasing safety risks in dense operations [56]. Tracking and conflict detection effectiveness analyses show that surveillance performance and airspace structure jointly determine whether conflicts are detected in time [54]. SoS simulation studies of UAM ecosystems show that disruptions (e.g., off-nominal passenger or vertiport events) can cascade into delays, diversions, cancellations, and network instability—demonstrating that operational resilience is an SoS property rather than a single-vehicle attribute [79,84,98].

3.3. Governance and evolution principle

3.3.1. New characteristics

From the SoS architecture aspect for governance and evolution principle, multiple stakeholders (e.g., operators, platform providers, UTM services, regulatory bodies) may pursue different goals under different policies. Supporting literature indicates that the LA SoS governance-evolution setting departs from traditional civil aviation in three aligned ways.

- (1) LA SoS governance expands to a multi-stakeholder regime with shared authority and competing objectives. Unlike conventional air traffic management, LA SoS often needs to coordinate a broader set of actors, including aircraft and fleet operators, vertiport and infrastructure owners, digital service providers, and public-interest stakeholders concerned with noise, privacy, equity, and community acceptance [73,77,78,83,88]. This broader governance surface means that system-level safety and performance cannot be ensured solely by operational procedures. Instead, governance choices directly shape what architectures are

viable, what constraints are binding, and how responsibilities are distributed as density scales [97].

- (2) LA SoS policies increasingly take effect through data-sharing and machine-enforceable rule ecosystems. In LA SoS, “policy” is not only written guidance but is operationalized through machine-enforceable constructs and information-sharing infrastructure. Risk-informed airspace constraints (e.g., population-aware restrictions and structured urban representations) translate societal exposure concerns into enforceable operational boundaries [25, 35]. At the same time, compliance and coordination depend on data rights and interoperable information flows, such as remote identification ecosystems, privacy-limited information sharing, and access assignment mechanisms that effectively become part of the operational rule layer [31,36,67]. As a result, governance and architecture co-evolve. Policy decisions set who can share what data and under what assurances, and these choices shape achievable outcomes for safety, efficiency, and accountability [77,78].
- (3) Rapid LA SoS evolution demands evidence-centered, continuously updated safety assurance. Compared with the slower certification and procedure cycles in traditional civil aviation, LA SoS evolves faster in autonomy, AI-enabled traffic services, and digital infrastructure. Governance therefore needs to sustain auditable safety arguments while accommodating frequent changes. This is reflected in digital-centered assurance and certification-aligned methodologies, including certification by simulation with explicit credibility concerns [27], structured evidence integration for safety assessment [47], and safety-case oriented standards and monitoring expectations that emphasize lifecycle assurance rather than one-time approval [38,99].

3.3.2. Risk patterns

We address risks associated with multi-stakeholder settings (e.g., operators, platform providers, UTM services, regulatory bodies), where different goals and policies may lead to inconsistent safety prioritization, misaligned incentives, and fragmented authority during critical events. Furthermore, asynchronous evolution and efficiency-driven local changes may induce new governance mismatch, rule interface inconsistency, unclear accountability, and safety drift when they are not aligned with SoS-level safety constraints. The reviewed literature supports these risk patterns as follows.

- (1) Divergent stakeholder objectives can cause uneven safety prioritization and gradual safety drift in LA SoS. Because LA SoS typically involves trade-offs among safety, capacity, cost, sustainability, and community constraints, stakeholders can weight objectives differently, producing uneven safety prioritization and gradual drift away from SoS-level constraints. Enterprise-architecture and safety-culture perspectives explicitly warn that dispersed responsibilities and differing priorities undermine shared learning loops and coordinated safety action [83]. Infrastructure-centered reviews (e.g., vertiport design and operations) further imply that mismatches among rules, design assumptions, and operational realities can create bottlenecks and compress safety margins as demand scales [73]. Lifecycle- and sustainability-oriented SoS analyses suggest that evolving energy mixes and policy expectations can make previously acceptable local operating choices inconsistent with wider sustainability targets, turning governance misalignment into an evolution risk [92].
- (2) Incentive misalignment can push stakeholders to optimize local efficiency at the expense of system resilience. Several SoS studies demonstrate that local optimizations (fleet strategy, booking rules, vertiport operations) can amplify congestion, delays, diversions, and network instability when coupled across the ecosystem, creating incentives to chase throughput while under-

investing in resilience [79,84]. Policy sensitivity analyses indicate that UAM performance can be highly sensitive to traffic and fleet management policies, meaning small rule or incentive changes can strongly reshape system-level outcomes and risk exposure [98]. Operational-limits studies and cross-city comparisons likewise show that hard constraints (throughput, weather, emissions, infrastructure) can turn demand shocks into systemic service degradation if governance and investment incentives are not aligned ahead of scale-up [86,97].

- (3) Fragmented authority can blur accountability and yield inconsistent responses during critical events in LA SoS. Governance-oriented reviews stress that without aligned regulatory frameworks and certifiable assurance processes for AI-heavy traffic services, rare or untrained scenarios can trigger systemic risk [38, 99]. Stress-testing and digital-twin works highlight that apparently compliant systems can remain brittle at the edges unless governance requires systematic probing and shared evidence practices [50,95]. Standardization-facing and rule-encoding work (e.g., digital traffic lights, risk-threshold separation standards, rule-constrained maneuver coordination, and conflict-management architectures) suggests that inconsistent rule implementation across services can turn fragmented authority into divergent tactical behavior and safety outcomes [46, 49,51,55,62].

4. Gap analysis

For the analysis of SoS-level risk patterns, this section focuses on the gaps between existing safety research and LA SoS safety. The existing SoS literature places substantial emphasis on reliability and resilience [132], focusing primarily on (i) SoS reliability modeling and evaluation, (ii) network reliability within SoS, (iii) recovery mechanisms/resilience dynamics, (iv) reliability-oriented SoS design and simulation/verification, and (v) reliability and evolution management in SoS engineering. Explicit SoS safety studies remain limited. Existing aviation safety research was largely developed under traditional aviation SoS architectures, and primarily address system-level risks (human, vehicle, CNS, or operational management) with clear system boundaries. As a new transportation SoS, LA SoS may have new emergent risk types and cross-domain cascading risk propagation via new architecture. This gap is amplified by LA safety and AI requirement. For safety, compared with traditional aviation, NASA’s UAM market study forecasts up to 11 million LA flights per day in the United States by 2030 [5]. At the current accident rate of 1.13 per million flights [133], this would imply about 12.4 LA accidents per day. This poses critical safety challenge for the existing aviation SoS architecture. For AI, NASA defines UAM Maturity Levels to map the staged development of a UAM transportation system [134]. At mid-to-high maturity (e.g., UAM Maturity Level-4), some functions must run without continuous human supervision to support hundreds of simultaneous operations within a single metropolitan area. This has gone beyond the existing safety management technology and in current aviation architecture. As pointed out, early design (architecture) is critical for ensuring safety, as up to 80% of safety-impacting decisions are made during the conceptual development phase [8]. Therefore, we analyze possible new challenges through architecture design, testing & evaluation, and safety management.

4.1. SoS safety-control-oriented architecture design

Existing SoS studies at the architectural level mainly synthesize key perspectives for architecture selection and evolution (e.g., interdependence, autonomy, uncertainty, and dynamic evolution), and provide surveys and toolkits to support complex SoS and cyber-physical system analysis and design [100,103,105,124]. Other works emphasize high-level governance principles but largely remain at the level of conceptual frameworks or visions [101,107,113,117]. In LA SoS

context, there is still a lack of LA SoS architecture design that enable fast reliable operational risk control via effective human-automation coordination under high-density, distributed, time-critical operations. In response, our architecture design introduces human-automation safety control loop as the basic unit, which embeds trustable AI to assure reliability, and proposes digital flight rules to enable coordinated LA safety control.

4.2. SoS safety testing & system-level safety constraints

The reviewed literatures offer a rich set of methods for modeling interdependent risks and systemic risk mechanisms, including SoS risk driven by shared states, decisions, and resources [108,109,120,122], systemic risk propagation under multi-objective decision-making and interdependent couplings [110,114], dependent-failure modeling [102], data-driven critical component identification [104] and techniques such as precursor analysis [111], dependability modeling [112], Bayesian networks [115], and multi-agent simulation [119] to reveal coincident and interacting hazards. These efforts mainly focus on safety model and analysis under external events [116,118,123], which remain difficult to offer traceable design-to-verification for SoS safety requirement and decompose it into system-level safety constraints. In response, LA SoS testing & evaluation needs validate the safety boundary and requirements under various interactions with environment in digital-physical combined test platform.

4.3. Real-time SoS safety management research

The IASMS concept of operations integrates safety assurance and risk management that together identify and mitigate emergent risks and hazards much more rapidly in-time than today, using less labor, and able to scale for increased complexity [135]. Studies on operational safety management make progress including, focusing on assessment, planning, and recovery—for example, resilience assessment and risk monitoring dashboards [106,113], precursor-based monitoring and decision support [111], and systemic disaster recovery and strategic preparedness [120,121]. For LA SoS safety management, it needs more systematic integration and development of real-time safety management towards an in-time, federated monitor-assess-mitigate architecture that fuses cross-domain data to predict risk and trigger mitigation in time.

5. LA SoS safety

5.1. Basic safety control loop for LA SoS

As Leveson's Systems Theoretic Accident Model and Process

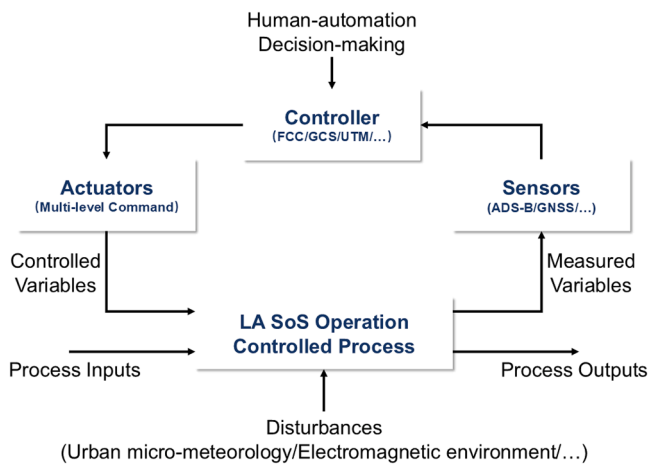


Fig. 3. Safety control loop for LA SoS.

(STAMP) theory defined [136], safety is not explained as a linear cause-effect chain but maintained through a closed-loop control structure: a controller observes the controlled process through sensors and keeps the process within safe limits. Following this concept, we define specific safety control loop (closed-loop) in LA SoS through real-time human-automation collaboration. Following Nancy Leveson's control loop [136], we have drawn Fig. 3 as an explicit “sensors-controller-actuators” control loop. In LA environments, safety is not achieved through a one-shot mitigation chain, but continuously maintained through a distributed human-automation closed-loop structure. Multiple controllers (e.g., Flight Control Computer (FCC)/Ground Control Station (GCS)/UTM) acquire multi-source sensing and feedback (e.g., Automatic Dependent Surveillance-Broadcast (ADS-B)/Global Navigation Satellite System (GNSS)/urban surveillance) to update the operational risk picture and issue multi-level commands to the corresponding actuators. This closed-loop enables real-time iterative “sense-control-act” cycles to cope with residual or emerging risks under LA conditions characterized by high dynamics, environmental disturbances, multi-actor interactions, and limited safety margins.

5.2. SoS safety engineering framework

Building on the safety control loop for LA SoS safety, we propose three safety engineering modules—architecture design, testing & evaluation, and safety management—jointly respond to risks analyzed above.

5.2.1. Architecture design

For organizational and operational structure risks, as LA transportation is shifting beyond the conventional aviation SoS architecture, it needs to design a new LA SoS architecture with reliable human-automation safety control and digital flight rule. Human-automation safety control loop is the basic unit of LA SoS safety, which decides how fast and reliable an operational risk can be controlled. Trustable AI will be deeply embedded to assure its reliability. Digital flight rules are a set of automated regulations which will coordinate LA traffic to maintain high safety and efficiency. Accordingly, this section categorizes LA SoS safety architectures into basic control, layered control, and risk-adaptive control designs.

Basic control architecture: The basic control architecture primarily targets risk control in low-density operational scenarios. In low-density LA operations where different safety control loops remain largely independent, the key is to keep coordination lightweight while preserving local autonomy. This can be done by standardizing a minimal cross-loop safety interface (e.g., a common format for sharing intent, contingency status, and safety margins) to reduce coordination overhead across independent operators and services. Meanwhile, the architecture should support event-triggered, time-bounded authority handover, so that a supervisory safety service can temporarily override an aircraft's local avoidance decision only under predefined emergency conditions. Existing SoS safety work on interface standardization and service-continuity dependability under stable conditions provides partial support [108,109], but it rarely addresses the second-level coupling and faster-switching required in LA operations.

Layered control architecture: The layered control architecture corresponds to higher safety requirements (i.e., multi-level risk management is needed to prevent cross-level cascading failures) and moderate environmental uncertainty (i.e., environmental fluctuations exist but are relatively predictable). It focuses on controlling risk propagation across hierarchical levels. For example, it may adopt a hierarchical distributed structure that sets risk-control measures at the basic network layer (physical units), the collaborative layer (cross-system resource integration), and the decision-making layer (global optimization) [101]. This idea is consistent with resilience- and dependability-oriented SoS frameworks that stress multi-layer protection and the structure of interdependencies [106,112,113]. However, these studies are seldom

instantiated for controlling LA operations, where the boundaries between layers themselves may need to be re-partitioned in real time.

Risk-adaptive control architecture: The risk-adaptive architecture corresponds to the highest safety requirements (i.e., responding to unprecedented risks) and high environmental uncertainty (i.e., dynamic environmental changes, frequent sudden disturbances, and unpredictable risks). It focuses on enhancing system resilience and adaptability. For example, it may establish a dynamic decision-making framework to adjust interactions between system modules, leverage bio-inspired heuristics to mitigate unprecedented risks, and realize real-time optimization of risk-control strategies. Recent concepts such as capability-oriented SoS safety and AI-enabled adaptive safety management point in this direction [101,120]. However, they offer limited concrete guidance on how to embed risk-adaptive safety loops into LA SoS architectures.

5.2.2. Testing & evaluation

For composition and environment interaction risks, we handle composition safety through SoS safety testing and system safety requirement identification. We derive SoS-level safety requirements and decompose them into system-level constraints (including airworthiness requirements) to ensure composition safety. In testing & evaluation, we validate these requirements and safety interactions with environment in digital-physical combined test platform via scenario-based evaluation.

Test Scenario Generation: Test scenarios take the designed SoS and safety architecture as inputs, making explicit how safety loops are composed, triggered and coupled across levels. Existing research provides useful elements: shared-state risk-modeling supports external-event-internal-response-cascading-failure scenarios [108–110,118], precursor analysis structures scenarios around degradation and recovery [111,118]. Multi-agent hazard analysis (e.g. SimHAZAN) perturbs node behavior to uncover accident paths beyond static techniques [119]. These methods remain tied to specific infrastructure and have not yet been extended into LA-oriented, cross-scale scenario libraries that co-encode LA characteristics and possible risk patterns should be studied in the future.

Evaluation Metric Construction: In our framework, metrics fall into three categories. Safety-loop process metrics quantify individual loops in terms of timeliness, accuracy and concurrency—for example, delays from hazard onset to detection, decision and response, and interactions between multiple simultaneously triggered loops [106,113,119]. At the network level, safety-loop metrics can leverage critical-component identification, dependent-failure modeling, and SoS dependability analysis to pinpoint nodes and links whose failure could trigger system-wide effects [102,104,112]. SoS-level metrics provide global measures of safety, reliability and resilience [101,113], and lack explicit mappings between “loop-network-SoS” layers and operational choices. A key task is therefore to build a metric system that is both traceable to top-level safety capabilities and decomposable into system-level safety requirements.

SoS Testing Environment: For LA SoS, large-scale physical experiments are expensive. SoS safety studies suggest several directions: multi-model and hierarchical simulations combining different fidelities across component, system and SoS levels [24,105,118,120], state-based modeling (e.g. Phantom System Models) that represent multiple subsystems and their shared states within a unified virtual system [72,74,118], and Monte-Carlo multi-agent simulations to evaluate safety and recovery of critical plants embedded in interdependent infrastructure [65,68,116]. These point toward a possible integrated LA testbed that combines digital twins, multi-agent models and multi-scale simulation, which needs to introduce real systems via hardware-in-the-loop into SoS testing and inject extreme perturbations.

5.2.3. Safety management

For governance and evolution principle risks, the framework extends traditional organization-centric SMS towards IASMS. Traditional SMS is

largely post-event and labor-intensive, which makes it difficult to detect and mitigate evolving risks. In contrast, IASMS extends SMS by integrating in-time safety assurance with traditional risk management experience and provides a distributed, federated “monitor-assess-mitigate” decision loop that fuses cross-domain data to predict risk and trigger mitigation in time.

Risk perception and decision: In high-density, highly uncertain LA environments, pre-defined rules and mission-level planning alone cannot cover all tail scenarios. Several technical directions already contribute to real-time risk perception and predictive situational awareness: urban flow simulations for wind and ground risk [21,25], digital LA airspace models for risk mapping and capacity assessment [35,41,42,63], dynamic and cascade-failure models for tracking evolving risks under real-time disturbances [107,111], and the use of large language models, knowledge graphs and graph-based retrieval for accident analysis and information integration [37,100]. Within an IASMS context, these approaches strengthen the “monitor–assess–predict” phases, yet they are rarely embedded as operational elements that can trigger “decide–mitigate” actions (e.g., rule updates or mission reconfiguration). This missing link, together with model uncertainty and data dependence, limits deployment and calls for tighter constraints, auditability, and traceability in the SoS management architecture.

Task-oriented collaboration: The task-oriented collaboration layer addresses how different safety agents should collaborate during federated “monitor-assess-mitigate” decision loop. Some work adopts centralized, on-ground tactical resolution and then broadcasts distributed speed/altitude instructions. It also benchmarks learning-enabled solvers against collision-avoidance standards, supporting auditability in an IASMS-like loop [34]. Other studies prioritize decentralized on-board self-separation. They use online decision-making with probabilistic safety constraints, so aircraft can mitigate conflicts under uncertainty [72,76]. For structured urban airspace, learning-based deconfliction integrates explicit conflict geometry and multi-objective costs. This helps scale coordination while keeping route/waypoint compliance [28,58]. Existing studies still lack a unified task-oriented collaboration view that specifies shared objectives, interoperable coordination primitives, and verifiable trigger-and-handover rules across these components.

Management Rules and Standards: From a safety-management transformation perspective, current work on UAM, U-space/UTM and multi-domain traffic management that defines management rules, service capabilities and staged operational levels [77–79], together with SoS safety studies advocating “capability-oriented, stepwise” safety management [119,122], can be viewed as early transitions from organization-centric SMS toward IASMS-like governance. In this view, rules and standards start to encode not only operational constraints but also safety capabilities and performance expectations. However, existing contributions remain largely conceptual, lacking explicit traceability between rules, safety control loops and local operational behavior, and offering limited insight on how a multi-operator LA SoS maintains a common safety baseline while allowing bounded differentiation.

6. Conclusions

In this paper, we consider the LA transportation system as a new SoS, and analyze new characteristics and risk patterns from the SoS architecture perspective. Specifically, existing aviation safety research was largely developed under traditional aviation SoS architectures, and primarily address system-level risks (human, vehicle, CNS, or operational management) with clear system boundaries. As a new SoS, LA SoS may have new emergent risk patterns and cross-domain cascading risk propagation via new architecture. Therefore, we propose LA SoS safety engineering framework to handle possible new challenges through architecture design, testing & evaluation, and safety management. This work does not aim to exhaustively cover these risk categories or provide

a comprehensive overview, but rather focuses on SoS-level risk patterns and safety engineering framework. Based on the critical synthesis, we propose the following insights and recommendations for future research:

- (1) **Architecture under human-automation collaboration and dynamic function allocation.** A key future direction is to make human-automation function allocation a first-class architectural concern for LA SoS safety, especially as enabling technologies (distributed sensing, connectivity, data platforms) and AI-based autonomy/decision support become deeply embedded. This implies safety-control architectures that explicitly place perception, decision authority, and intervention capability across onboard AI, ground automation services, and human supervisors, and that can reconfigure when density increases, model uncertainty rises, or communications degrade.
- (2) **Scenario data-driven testing for rare hazards.** Future research should strengthen scenario ecosystems and data assimilation pipelines that reflect both physical and decision uncertainties. Testing and assessment should explicitly target low-frequency, high-impact events and their cross-domain risk chains, including AI brittleness. SoS unified testing approach that blends high-fidelity simulation, uncertainty-aware modeling, and continuous field-data updates can help reveal previously unseen hazard combinations and quantify how learning errors propagate into coordination breakdowns and safety-margin erosion.
- (3) **In-time safety governance under rapid evolution.** Another promising direction is operationalizing in-time safety assurance in an ecosystem where models evolve continuously through software updates, model retraining, and changing data. This raises challenges that an updated learning model should remain compatible with digital flight rules and cross-operator coordination. Future work should explore governance mechanisms that couple real-time learning with auditable change management, so that AI-enabled services can be deployed and adapted without weakening system-wide safety alignment.

CRediT authorship contribution statement

Anzhao Yao: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Shanghan Li:** Writing – review & editing, Writing – original draft, Investigation. **Kaifeng Feng:** Writing – review & editing, Writing – original draft, Investigation. **Tengfei Zhang:** Writing – review & editing, Writing – original draft, Investigation. **Xueying Song:** Writing – review & editing, Writing – original draft, Investigation. **Lizhi Wang:** Writing – review & editing, Writing – original draft, Investigation. **Ruixin Wang:** Writing – original draft, Investigation. **Peng He:** Writing – review & editing, Writing – original draft, Investigation. **Hang Zhou:** Writing – original draft, Investigation. **Hang Li:** Writing – review & editing, Investigation. **Shuiting Ding:** Writing – review & editing, Conceptualization. **Daqing Li:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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